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INSIDE: LLMs' INTERNAL STATES RETAIN THE POWER OF HALLUCINATION DETECTION

Chao Chen[1], **Kai Liu**[2], **Ze Chen**[1], **Yi Gu**[1], **Yue Wu**[1], **Mingyuan Tao**[1]
Zhihang Fu[1*], **Jieping Ye**[1*]
[1]Alibaba Cloud [2]Zhejiang University
{ercong.cc, zhihang.fzh, yejieping}@alibaba-inc.com

ABSTRACT

Knowledge hallucination have raised widespread concerns for the security and reliability of deployed LLMs. Previous efforts in detecting hallucinations have been employed at logit-level uncertainty estimation or language-level self-consistency evaluation, where the semantic information is inevitably lost during the token-decoding procedure. Thus, we propose to explore the dense semantic information retained within LLMs' **IN**ternal **S**tates for hallu**c**I**n**ation **DE**tection (**INSIDE**). In particular, a simple yet effective **EigenScore** metric is proposed to better evaluate responses' self-consistency, which exploits the eigenvalues of responses' covariance matrix to measure the semantic consistency/diversity in the dense embedding space. Furthermore, from the perspective of self-consistent hallucination detection, a test time feature clipping approach is explored to truncate extreme activations in the internal states, which reduces overconfident generations and potentially benefits the detection of overconfident hallucinations. Extensive experiments and ablation studies are performed on several popular LLMs and question-answering (QA) benchmarks, showing the effectiveness of our proposal. [1]

1 INTRODUCTION

Large Language Models (LLMs) have recently achieved a milestone breakthrough and demonstrated impressive abilities in various applications (Ouyang et al., 2022; OpenAI, 2023). However, it has been widely observed that even the state-of-the-art LLMs often make factually incorrect or nonsense generations (Cohen et al., 2023; Ren et al., 2022; Kuhn et al., 2022), which is also known as knowledge hallucination (Ji et al., 2023). The potentially unreliable generations make it risky to deploy LLMs in practical scenarios. Therefore, hallucination detection, that is, accurately detecting and rejecting responses when hallucinations occur in LLMs, has attracted more and more attention from the academic community (Azaria & Mitchell, 2023; Ren et al., 2022; Kuhn et al., 2022).

The token-level uncertainty estimation (e.g., predictive confidence or entropy) has shown its efficacy in hallucination detection on conventional NLP tasks (Malinin & Gales, 2020; Huang et al., 2023). However, how to derive the sentence-level uncertainty from the token-level remains a challenge, especially for modern auto-regressive LLMs whose response contents are generally diverse and sophisticated (Malinin & Gales, 2020; Kuhn et al., 2022; Duan et al., 2023). Thus, to avoid complicated token-to-sentence uncertainty derivation, researchers propose to evaluate the sentence uncertainty by the output languages directly (Kadavath et al., 2022; Yin et al., 2023; Zhou et al., 2023). Among the recent advancements, prompting LLMs to generate multiple responses to the same question and evaluating the *self-consistency* of those responses has been proven effective in hallucination detection (Wang et al., 2022; Shi et al., 2022). However, such a post-hoc semantic measurement on decoded language sentences is inferior to precisely modeling the logical consistency/divergence Manakul et al. (2023); Zhang et al. (2023).

Hence, instead of logit-level or language-level uncertainty estimation, this paper proposes to leverage the internal states of LLMs to conduct hallucination detection. The motivation is intuitive:

*Corresponding Author
[1]Code is available at https://github.com/alibaba/eigenscore

LLMs preserve the highly-concentrated semantic information of the entire sentence within their internal states (Azaria & Mitchell, 2023), allowing for the direct detection of hallucinated responses in the sentence embedding space.

In particular, with the generalized framework of **IN**ternal **S**tates for hallucInation **DE**tection (**INSIDE**), this paper performs hallucination detection from two perspectives. First, skipping secondary semantic extraction via extra models, we directly measure the self-consistency/divergence of the output sentences using internal states of LLMs. In order to explore semantic consistency in the embedding space, Section 3.1 introduces an **EigenScore** metric regarding the eigenvalues of sentence embeddings' covariance matrix. Second, to handle the self-consistent (overconfident) hallucinations, we propose to rectify abnormal activations of the internal states. Specifically, Section 3.2 develops a feature clipping approach to truncate extreme features, which tends to prevent overconfident generations during the auto-regressive procedure. In Section 4, the effectiveness of our method is validated through extensive experiments on several well-established QA benchmarks.

The main contributions of our work are as follows:

- We propose a generalized INSIDE framework that leverages the internal states of LLMs to perform hallucination detection.
- We develop an EigenScore metric to measure the semantic consistency in the embedding space, and demonstrate that the proposed EigenScore represents the differential entropy in the sentence embedding space.
- A test time feature clipping approach is introduced to truncate extreme activations in the feature space, which implicitly reduces overconfident generations and helps identify the overconfident hallucinations.
- We achieve state-of-the-art hallucination detection performance on several QA benchmarks, and conduct extensive ablation studies to verify the efficacy of our method.

2 Background on Hallucination Detection

In this work, we mainly focus on the knowledge hallucination detection of natural language generation based on LLMs, especially for Q&A task (Reddy et al., 2019; Kwiatkowski et al., 2019). Given an input context $\boldsymbol{x}$, a typical LLM (Zhang et al., 2022; Touvron et al., 2023a) parameterized with $\boldsymbol{\theta}$ is able to generate output sequences in autoregressive manner $y_t = f(\boldsymbol{x}, y_1, y_2, \cdots, y_{t-1}|\boldsymbol{\theta})$, where $\boldsymbol{y} = [y_1, y_2, \cdots, y_T]$ denotes the output sequence and y_t denotes the t-th output token. We denote $p(y_t|y_{<t}, \boldsymbol{x})$ the Maximum Softmax Probability (MSP) of t-th token. For a traditional classification model, the MSP measures the confidence level of the classification result and has been widely used as an uncertainty measure of predictions (Hendrycks & Gimpel, 2016). Therefore, for sequence generation task, a straightforward sequence uncertainty can be defined as the joint probability of different tokens, which is known as **Perplexity** (Ren et al., 2022),

$$P(\boldsymbol{y}|\boldsymbol{x}, \boldsymbol{\theta}) = -\frac{1}{T} \log \prod_t p(y_t|y_{<t}, \boldsymbol{x}) = -\frac{1}{T} \sum_t \log p(y_t|y_{<t}, \boldsymbol{x}) \tag{1}$$

As shorter sequences generally have lower perplexity, the length of the output sequence T is utilized to normalize the joint probability. Since different tokens contribute differently to the semantics of the sentence (Raj et al., 2023; Duan et al., 2023), the perplexity defined by averaging token-level uncertainty cannot effectively capture the uncertainty of the entire sequence. It has been demonstrated that utilizing multiple generations for one input is beneficial to estimate the sequence-level uncertainty (Malinin & Gales, 2020; Kuhn et al., 2022; Manakul et al., 2023). We denote $\mathcal{Y} = [\boldsymbol{y}^1, \boldsymbol{y}^2, \cdots, \boldsymbol{y}^K]$ as K generated responses for input context $\boldsymbol{x}$. For a given LLM, multiple responses could be easily obtained by the top-p/top-k sampling strategy during inference time (Touvron et al., 2023a; Kadavath et al., 2022). In Malinin & Gales (2020), the **Length Normalized Entropy** is proposed to measure the sequence-level uncertainty by making use of multiple generations, which is defined as

$$H(\mathcal{Y}|\boldsymbol{x}, \boldsymbol{\theta}) = -\mathbb{E}_{\boldsymbol{y} \in \mathcal{Y}} \frac{1}{T_{\boldsymbol{y}}} \sum_t \log p(y_t|y_{<t}, \boldsymbol{x}) \tag{2}$$

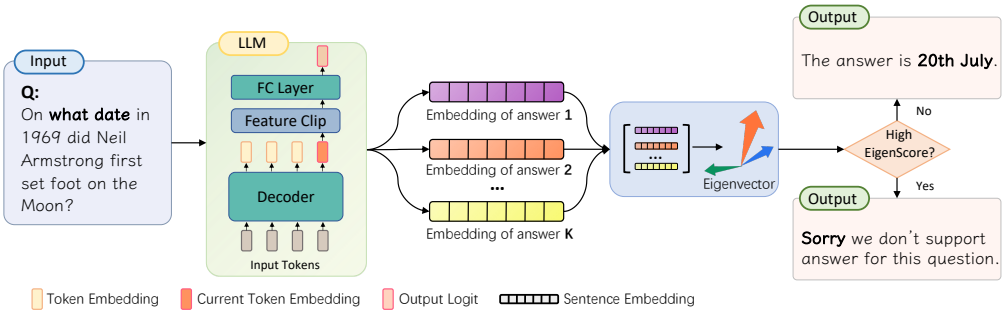

Figure 1: Illustration of our proposed hallucination detection pipeline. During inference time, for a given question, the extreme features in the penultimate layer are truncated and the EigenScore is computed based on the sentence embeddings across multiple responses.

When a model is uncertain about its response, it generates hallucination context, resulting in an answer distribution with a high entropy (Kadavath et al., 2022). It has been shown that the length-normalized entropy performs better than the non-normalized one (Lin et al., 2023).

In addition to the predictive uncertainty or entropy, the semantic consistency (Lin et al., 2023; Raj et al., 2023) among multiple responses has also been widely explored to measure the hallucination degree of LLMs, which hypothesis that the LLMs are expected to generate similar outputs if they know the input context and they are sure about the answers (Wang et al., 2022; Manakul et al., 2023). An intuitive semantic consistency metric is **Lexical Similarity** (Lin et al., 2022; 2023), which explores the average similarity across multiple answers as consistency measure

$$S(\mathcal{Y}|\boldsymbol{x},\boldsymbol{\theta}) = \frac{1}{C}\sum_{i=1}^{K}\sum_{j=i+1}^{K} sim(\boldsymbol{y}^i,\boldsymbol{y}^j) \tag{3}$$

where $C = K \cdot (K-1)/2$ and $sim(\cdot,\cdot)$ is the similarity defined by Rouge-L Lin (2004).

3 Method

In this section, we introduce the details of our proposed INSIDE framework for hallucination detection. The whole pipeline is illustrated as Fig. 1. In section 3.1, we demonstrate a simple but effective EigenScore metric by exploring sentence-level semantics in the internal states of LLMs. In section 3.2, a test-time feature clipping approach is introduced to effectively alleviate the issue of overconfident generation, thereby aiding in the identification of self-consistent hallucinations

3.1 Hallucination Detection by EigenScore

The existing uncertainty or consistency based hallucination detection metrics are exploited in the logit or language space, which neglect the dense semantic information that is retained within the internal states of LLMs. To better exploit the dense semantic information, we propose to measure the semantic divergence in the sentence embedding space. For the t-th output token y_t, we denote the hidden embedding in the l-th layer as $\boldsymbol{h}_t^l \in \mathbb{R}^d$, where d is the dimension of the hidden embedding ($d = 4096$ for LLaMA-7B and $d = 5120$ for LLaMA-13B). According to Ren et al. (2022); Azaria & Mitchell (2023), the sentence embedding can be obtained by averaging the token embedding $\boldsymbol{z} = \frac{1}{T}\sum_{t=1}^{T}\boldsymbol{h}_t$, or taking the last token embedding as sentence embedding $\boldsymbol{z} = \boldsymbol{h}_T$. In our main experiments, we use the embedding of the last token in the middle layer as the sentence embedding, as it effectively captures the sentence semantic (Azaria & Mitchell, 2023). The comparison results of using different sentence embeddings are demonstrated in the ablation studies 4.3. For K generated sequences, the covariance matrix of K sentence embeddings can be computed as

$$\boldsymbol{\Sigma} = \mathbf{Z}^\top \cdot \mathbf{J}_d \cdot \mathbf{Z} \tag{4}$$